

Normalisation of two forms

Weekly running sheet - Normalisation

UNF

BOOKING(cust_id, cust_name, cust_business, room_id, room_name, book_start_datetime, book_end_datetime, book_n_guest, base_rate_guest, room_rate_hour, cuisine_type, cuisine_rate_guest, book_total_cost, decor_theme_name)

1NF

BOOKING(cust_id, book_start_datetime, cust_name, cust_business, room_id, room_name, book_end_datetime, book_n_guest, base_rate_guest, room_rate_hour, cuisine_type, cuisine_rate_guest, book_total_cost, decor_theme_name)

Candidate keys:

1. cust_id, book_start_datetime
2. room_id, book_start_datetime

Partial Dependencies:

- cust_id -> cust_name, cust_business
- room_id -> room_name, room_rate_hour

2NF

BOOKING(cust_id, book_start_datetime, room_id, book_end_datetime, book_n_guest, base_rate_guest, cuisine_type, cuisine_rate_guest, book_total_cost, decor_theme_name)

CUSTOMER(cust_id, cust_name, cust_business)

ROOM(room_id, room_name, room_rate_hour)

Transitive Dependencies:

- cuisine_type -> cuisine_rate_guest

3NF

BOOKING(cust_id, book_start_datetime, room_id, book_end_datetime, book_n_guest, base_rate_guest, cuisine_type, book_total_cost, decor_theme_name)

CUSTOMER(cust_id, cust_name, cust_business)

ROOM(room_id, room_name, room_rate_hour)

CUISINE(cuisine_type, cuisine_rate_guest)

Full-dependency

cust_id, book_start_datetime -> room_id, book_end_datetime, book_n_guest,
base_rate_guest, cuisine_type, book_total_cost, decor_theme_name

cust_id -> cust_name, cust_business

room_id -> room_name, room_rate_hour

cuisine_type -> cuisine_rate_guest

Function room evaluation - Normalisation

UNF

EVALUATION(eval_date, chef_id, chef_name, room_id, room_name, room_address,
room_capacity, room_parking, eval_approval, eval_comment)

1NF

EVALUATION(eval_date, room_id, chef_id, chef_name, room_name, room_address,
room_capacity, room_parking, eval_approval, eval_comment)

Candidate keys:

1. eval_date, room_id

Partial Dependencies:

room_id -> room_name, room_address, room_capacity, room_parking

2NF

EVALUATION(eval_date, room_id, chef_id, chef_name, eval_approval, eval_comment)

ROOM(room_id, room_name, room_address, room_capacity, room_parking)

Transitive Dependencies:

chef_id -> chef_name

3NF

EVALUATION(eval_date, room_id, chef_id, eval_approval, eval_comment)

ROOM(room_id, room_name, room_address, room_capacity, room_parking)

CHEF(chef_id, chef_name)

Full dependency

eval_date, room_id -> chef_id, eval_approval, eval_comment

room_id -> room_name, room_address, room_capacity, room_parking

chef_id -> chef_name

Final Combined Normalisation (both forms)

BOOKING(cust_id, book_start_datetime, room_id, book_end_datetime, book_n_guest, base_rate_guest, cuisine_type, book_total_cost, decor_theme_name)

CUSTOMER(cust_id, cust_name, cust_business)

ROOM(room_id, room_name, room_rate_hour, room_address, room_capacity, room_parking)

CUISINE(cuisine_type, cuisine_rate_guest)

EVALUATION(eval_date, room_id, chef_id, eval_approval, eval_comment)

CHEF(chef_id, chef_name)